Consider the equations $4x_1+x_2 = 40$ and $2x_1+3x_2 = 90$.

$4x_1+x_2 = 40$ is a line passing through the points (0,40) and (10,0). Any point lying on or above the line $4x_1+x_2= 40$ satisfies the constraint $4x_1+ x_2 \geq 40$.

$2x_1+3x_2 = 90$ is a line passing through the points (0,30) and (45,0). Any point lying on or above the line $2x_1+3x_2= 90$ satisfies the constraint $2x_1+3x_2 \geq 90$.

Draw the graph using the given constraints.

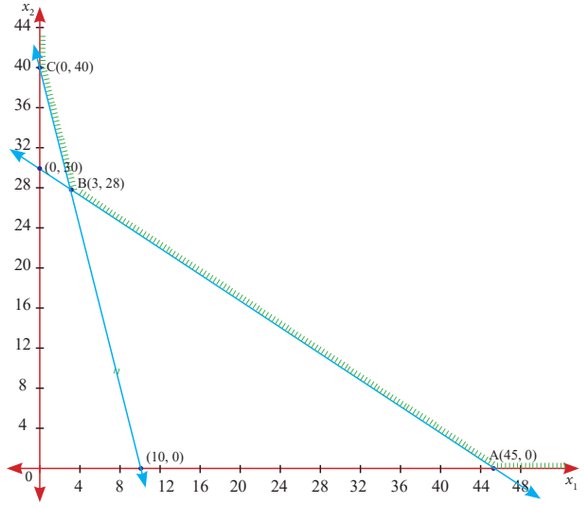



Fig 10.2

The feasible region is ABC (since the problem is of minimization type we are moving towards the origin.

| Corner points | $z = 5x_1+4x_2$ |
|---|---|
| A(45,0) | 225 |
| B(3,28) | 127 |
| C(0,40) | 160 |

Table 10.3

The minimum value of Z occurs at B(3,28).

Hence the optimal solution is $x_1 = 3$, $x_2 = 28$ and $Z_{min}$=127.

**Example 10.7**

Solve the following LPP.

Maximize $Z= 2x_1+3x_2$ subject to constraints $x_1 + x_2 \leq 30$ ; $x_2 \leq 12$; $x_1 \leq 20$ and $x_1, x_2 \geq 0$.

***Solution:***

We find the feasible region using the given conditions.

Since both the decision variables $x_1$ and $x_2$ are non-negative, the solution lies in the first quadrant of the plane.

Write all the inequalities of the constraints in the form of equations.

Therefore we have the lines

$x_1+x_2=30$; $x_2=12$; $x_1= 20$

$x_1+x_2=30$ is a line passing through the points (0,30) and (30,0)

$x_2 = 12$ is a line parallel to $x_1$–axis

$x_1 = 20$ is a line parallel to $x_2$–axis.

The feasible region satisfying all the conditions $x_1+ x_2\leq 30$; $x_2\leq 12$ ; $x_1\leq 20$ and $x_1, x_2 \geq 0$ is shown in the following graph.

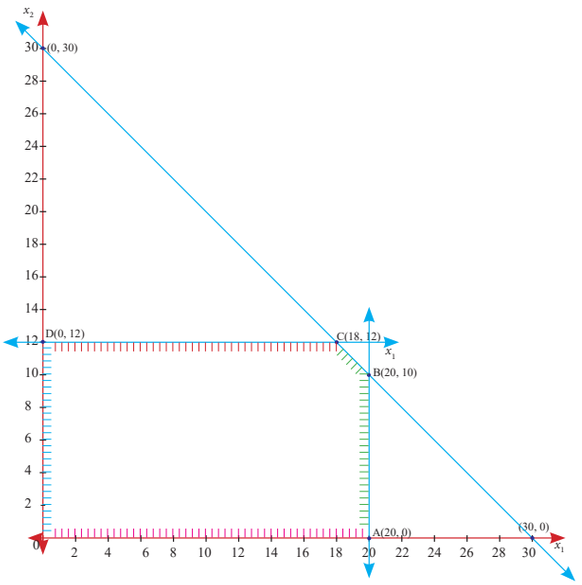



Fig 10.3



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